

Curriculum Vitae

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Professional Objective

To develop expertise in experimental Astroparticle Physics through cutting-edge research at world-class laboratories, while contributing to the advancement of ultra-high energy neutrino detection technologies.

Education

Ph.D. in Physics *2020–Present*

University of Kansas, USA

3.98/4.0

M.S. in Electrical Engineering *2024–2025*

University of Kansas, USA

3.87/4.0

M.Sc. in Physics *2018–2020*

Aligarh Muslim University, India

9.170/10 (Gold Medalist)

B.Sc. Honors in Physics *2015–2018*

Aligarh Muslim University, India

9.186/10 (Gold Medalist)

Higher Secondary (12th) *2014*

Central Board of Secondary Education, India

9.120/10 (First Position)

Secondary (10th) *2012*

Central Board of Secondary Education, India

10/10 (Certificate of Merit)

Research Experience

Primary Analyzer, Askaryan Radio Array (ARA) Station 1 *2021–Present*

Conducting data analysis of ARA Station 1 for ultra-high energy (UHE) neutrino detection with Prof. D. Besson at University of Kansas (KU).

EECS Master’s Project Research *2024–2025*

Developed **AAFIYA**: Antenna Analysis in Frequency-domain for Impedance and Yield Assessment, a modular Python toolkit for rigorous, reproducible analysis and visualization of antenna S-parameters, impedance, and gain/beam patterns from both measurement and simulation data, with Prof. Jim Stiles at EECS Department, KU, (Report), (Slides).

Machine Learning Project *2025*

Developed a CNN model for “Automatic Surface Detection in Echograms using Deep Learning” as a part of Machine Learning class at EECS Department, KU, Project Repository), (Report).

IceCube Neutrino Observatory Monitoring *2021–2025*

Served as KU representative for monitoring shifts, presenting weekly reports to the collaboration.

Physics Master’s Thesis Research *2019–2020*

Investigated light sterile neutrino oscillations with Prof. M.S. Athar at Aligarh Muslim University (AMU), producing review report “Light Sterile Neutrinos: Status and Prospects” (DOI:10.13140/RG.2.2.13777.66405).

Summer Research Program *May-Jul 2019*

Studied the role of θ_{13} mixing angle in three-flavor neutrino oscillations with Prof. S.K. Agarwalla at Institute of Physics, India, producing technical report “Importance of 1-3 Mixing Angle in Three-Flavor Neutrino Oscillations Paradigm” (DOI: 10.13140/RG.2.2.34421.12009).

Winter Internship Program *Dec 2019-Jan 2020*

Researched “Weak Interaction Physics” at IIT-BHU with Dr. G. Abbas.

Physics Undergraduate Project Research *2017-2018*

“Quarks & Leptons”: Detailed study of Standard Model interactions with Prof. M.S. Athar at AMU [View Handwritten Report].

Field Experience

NSF’s Summit Station, Greenland Field Campaign *Apr-May 2025*

Participated in the Radio Neutrino Observatory – Greenland (RNO-G) calibration campaign, performing in-situ radio-frequency measurements and equipment deployment at Summit Station, Greenland.

Publications

First-Author Works

- **Seikh, M.F.H.** (2025). “A geometrical approach to neutrino oscillation parameters”. *AIP Advances* 15 (10): 105204. DOI:10.1063/5.0292233.
- **Seikh, M.F.H., Grachev, V., Kravchenko, I., Athar, M.S., & Besson, D.** 2025. “First Radio Neutrino & Cosmic Rays Astronomy Workshop in India”. *European Physics Journal-Special Topics (EPJ-ST)*. arXiv:2509.20704 DOI:10.1140/epjs/s11734-025-01955-8.
- **Seikh, M.F.H., Giri, P. & Besson, D. for the ARA Collaboration** 2025. “Techniques for Continuous Wave Identification and Filtering in the Askaryan Radio Array”. *PoS(ICRC2025)1171*. arXiv:2509.20735.
- **Seikh, M.F.H. & Besson, D. for the RNO-G Collaboration** 2025. “Observation of Radio Solar Flares in RNO-G”. *PoS(ICRC2025)1365*.
- **Seikh, M.F.H.** (2024). “Physics of radio antennas”. *European Physics Journal-Special Topics (EPJ-ST)*. arXiv:2411.07507 DOI: 10.1140/epjs/s11734-025-01500-7
- **Seikh, M.F.H. for the ARA Collaboration** (2024). “Askaryan Radio Array: searching for the highest energy neutrinos”. *European Physics Journal-Special Topics (EPJ-ST)*. arXiv:2411.01761 DOI: 10.1140/epjs/s11734-025-01708-7
- **Seikh, M.F.H. & Besson, D.** (2024). “IceCube Neutrino Observatory: A Cold Discovery Hub”. *Page 14-23, Physics Bulletin, Physics Department, AMU*
- **Seikh, M.F.H. for the ARA Collaboration** (2023). “Calibration and Physics with ARA Station 1”. *PoS(ICRC2023)1163*. arXiv:2308.07292
- **Seikh, M.F.H.** (2020). “Neutrinos and the 40 Future Routes”. *Physics Bulletin, Physics Department, AMU*. DOI: 10.13140/RG.2.2.27981.10729

Collaborative Works

Full publication (includes contributions to ARA, RNO-G, ARIANNA, IceCube, IceCube Gen2, PUEO, TAROGE-M and RET collaborations) record available on:

Inspire HEP ; Google Scholar ; ResearchGate ; ORCiD  ; Scopus

Professional Service

Editorial

- **Book Editor** for Springer’s EPJ-ST journal: “Radio Detection of Ultra-high Energy Neutrino & Cosmic Ray” (15 articles) with M.S. Athar, D. Besson, V. Grachev, and I. Karvchenko (*2024–2025*)
- **Founder and Editor-in-Chief** of *The Magnetic Monopole* e-magazine for KU’s Society of Physics Students [View Issue] (*June 2022*)

Research Article Review

- **Peer Reviewer:** for World Scientific’s IJMP-A journal (1 article, 1 review) (*2025*)
- **Peer Reviewer:** for Springer’s EPJ-ST journal (5 articles, 11 reviews) (*2024–2025*)
- **Internal ARA Collaboration Reviewer:** Pawan Giri for ARA Collaboration, “A Comprehensive Diffuse Neutrino Search Using the Full Askaryan Radio Array”, 32nd International Symposium on Lepton Photon Interactions at High Energies (*November 2025*)

Professional Leadership/Membership

- **American Physical Society (APS):** Graduate Student Member (*2020–Present*)
- **National Society of Black Physicists:** Student Member (*2021–Present*)
- **Physics & Astronomy Graduate Student Organization (GSO), KU:**
 - ✓ Graduate Students Representative - Member at Large (*2025–Present*)
 - ✓ Graduate Students Representative - Liaison (*2024–2025*)
 - ✓ GSO Executive Officer (*2023–2024*)
- **Sir Syed Global Scholar Award (SSGSA):**
 - ✓ Peer Mentor (*2023–Present*)
 - ✓ Team Lead, Promotions & Outreach (*2023–2025*)
 - ✓ Member, Mentorship & Applications Team (*2021–2023*)
 - ✓ AMU Campus Team Member (*2019–2020*)
- **Society of Physics Students, KU Chapter:** Graduate President (*2021–2022*)
- **Sigma Pi Sigma (National Physics Honor Society):** Lifetime Member
- **AMU Society of Physics:** Member (*2018–Present*)

Technical Proficiencies

- **Programming Languages:** C/C++, Python, FORTRAN
- **Physics Software:** CERN ROOT, AraRoot, Wolfram Mathematica, HFSS
- **Operating Systems:** Linux (Ubuntu), Windows, macOS
- **Document Preparation:** L^AT_EX, Microsoft Office Suite

Honors & Awards

Academic Honors

- **University Gold Medals:**
 - ✓ M.Sc. Physics (Top among 50 students), AMU (*2020*)
 - ✓ B.Sc. Honors Physics (Top among 120 students), AMU (*2018*)
 - ✓ Highest percentage in B.Sc. (Hons) Faculty of Science, AMU (*2018*)

- **Abdul Aziz Medal** for B.Sc. Physics, AMU (2018)
- **Undergraduate Topper Scholarship**, UGC India (2020)
- **University Merit Financial Award**, AMU (2019–2020)
- **First Position** in Higher Secondary (12th) among 40 students (2014)
- **Certificate of Merit** for perfect 10/10 CGPA in Secondary (10th) (2012)

Fellowships & Grants

- **Ralston Dissertation Fellowship**, KU Physics (2025–2026)
- **McKeithan Summer Fellowship**, KU Physics (2025)
- **Graduate Student Award for Distinguished Service**, KU (2024–2025)
- **Wallace S. Strobel Scholarship**, KU EECS (2024–2025)
- **Ralston Summer Fellowship**, KU Physics (2024)
- **Doctoral Student Research Fund**, KU (2024)
- **Distinguished Students Award**, APS FIP (2021–2022)
- **Carl & Catherine Chaffee Scholarship**, KU Physics (2020–2021)

Conference Travel Awards

- **Graduate Student Travel Fund**, KU Physics (2023, 2025)
- **DAP Travel Grant**, APS (2024)
- **DPF Travel Award**, APS (2022, 2024)
- **CLAS Graduate Scholarly Development Fund**, KU (2022)

Other Distinctions

- **Washington DC Scholarship**, AMU Alumni Association (2015–2019)
- **Sir Syed Global Scholar Award** for prospective graduate students (2018)

Invited Lectures, Seminars, & Colloquia

- **Radio Engineering for Particle Astrophysics Experiments**
HEP Seminar, University of Nebraska-Lincoln (March 26, 2025)
- **Radio Engineering for Particle Astrophysics Experiments**
Astroparticle Seminar, Ohio State University (January 22, 2025)
- **Radio Engineering & Antenna Physics**
Lecture Series, IEI Summer School 2024, University of Alaska, Anchorage (July 2–5, 2024)
- **UHE Neutrino & Cosmic Ray Detectors**
ICEHAP Seminar, Chiba University, Japan (August 9, 2023)
- **Antenna Physics & Measurements**
Lecture Series, IEI Summer School 2023, South Dakota School of Mines (July 2–5, 2023)
- **UHE Neutrino & Cosmic Ray Detectors**
Physics Colloquium, AMU, India (January 12, 2023)
- **Antennas in UHE Neutrino Detectors**
Particle Physics Seminar, University of Delaware (October 4, 2022)

Outreach & Leadership

- Co-organized and hosted India’s inaugural workshop on **Radio Detection of UHE Neutrino and Cosmic Rays** at AMU (April 2024)

- Conducted outreach seminar on **Physics Research Opportunities** at AMU (2021)
- Participated in India's first **Science Leadership Workshop** (2020)
- Volunteer for **One Person One Tree** plantation drive (2017)
- Participant in **Blood Donation Camp** (2017)

Conference Presentations

Oral Presentations

- **RNO-G in-ice Antennas**
RNO-G Collaboration Meeting, Uppsala University (September 9–14, 2024)
- **Radio Frequency Instrumentation**
Radio Neutrino Workshop, AMU, India (April 18–20, 2024)
- **Towards Five Station ARA Analysis**
APS April Meeting, Sacramento (April 3–6, 2024)
- **ARA Antenna Models**
ARA Collaboration Meeting, Ohio State University (March 5–8, 2024)
- **Current Status & Future of ARA**
APS Prairie Section Meeting, University of Missouri (November 30–December 2, 2023)
- **Introduction to Antenna Physics**
RET Collaboration Meeting, Vrije University Brussels (November 8–11, 2023)
- **Gain Calibration of RNO-G Antennas**
RNO-G Fall Meeting, Penn State University (September 11–15, 2023)
- **Galactic Noise in Radio Neutrino Experiments**
Astro Seminar, KU (October 28, 2022)
- **Towards ARA Station 1 Calibration**
APS April Meeting, New York City (April 9–12, 2022)
- **RNO-G Ice Calibration Planning**
RNO-G Collaboration Meeting, University of Chicago (April 8, 2022)

Poster Presentations

- **CW Identification & Filtering Techniques in ARA**
39th ICRC, Geneva, Switzerland (July 19, 2025)
- **Observation of radio solar flares in RNO-G**
39th ICRC, Geneva, Switzerland (July 15, 2025)
- **Gain Measurements of LPDA Antennas**
APS April Meeting, Sacramento (April 3–6, 2024)
- **Overview of ARA Detector: Five Station Analysis**
ICTS Neutrino School, India (April 22–May 3, 2024)
- **Calibration of ARA Station 1**
38th ICRC, Nagoya University, Japan (July 29, 2023)

Professional Development

- **Responsible Conduct of Research**, (CITI Program, 2021, 2025) - 100%
- **From the Big Bang to Dark Energy** (U Tokyo Coursera, 2020) - 100%
- **Particle Physics: Introduction** (U Geneva Coursera, 2019) - 100%
- **Senior Diploma in Painting** (Bangiya Sangeet Parishad) - Distinction

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Special Topics

Radio Detection of Ultra-high Energy Neutrino and Cosmic Ray

Mohammad Ful Hossain Seikh, Mohammad Sajjad Athar,
Dave Besson, Victor Grachev, and Ilya Kravchenko (Eds.)



Cosmic rays and neutrinos: exploring high-energy messengers through radio detection.
Courtesy of Mohammad Ful Hossain Seikh

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